# Venkata Lakshmi Parimala Pasupuleti

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## SUMMARY

Senior software engineer with **9+ years shipping production systems** and deep hands-on ownership of GenAI agents, evaluation, and enterprise compliance. At Intuit, **builds and operates production Generative AI workflows** end to end: tool calling, orchestration, state management, RAG, structured outputs, and LLMOps instrumentation. At Entrata (100K+ MAU regulated SaaS), owned **PHI/HIPAA data infrastructure**, AWS production operations (S3, EC2, RDS, Lambda, IAM), Terraform infrastructure as code, and GitHub Actions CI/CD - shipping and operating production services, not editing templates. Built **Leasel**: an 8-agent LangGraph system with human-in-the-loop controls, provenance on AI outputs, and agent observability (93.6% precision across 1,240 test cases). Strong in Python, React, SQL, REST APIs, and the full AWS/Terraform/GitHub Actions stack for regulated environments. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## CORE COMPETENCIES

GenAI Agents & Workflow Orchestration | Tool Use / State Management / Tool Execution | Human-in-the-Loop & Agent Observability | LLM Evaluation & LLMOps | Python / React / SQL / REST APIs | AWS Production Operations (IAM, S3, EC2, RDS, Lambda) | Terraform / GitHub Actions CI/CD | PHI / HIPAA / Enterprise Compliance | Authentication & Authorization | Observability / Logging / Monitoring | Enterprise System Integrations & APIs | Root Cause Analysis & Production Debugging

## EXPERIENCE

### Intuit

Jun 2026 -- Present

**Software Engineer - GenAI & AI Systems** - Virginia, USA

- **Serves as technical point of contact for AI tool vendors and business stakeholders**: scopes integration requirements, evaluates third-party APIs, aligns on data contracts, and drives each initiative from technical discovery through go-live - working directly with TPMs and product managers to translate business needs into shipped solutions.
- **Builds and operates production Generative AI agents and workflows**: tool calling, orchestration with LangGraph, state management, RAG, context management, and structured outputs interacting with enterprise financial data systems.
- Implements **evaluation, observability, and human-in-the-loop controls**: LLM-as-judge harnesses, golden datasets, provenance on AI outputs; sets expectations with stakeholders on accuracy and edge cases before production rollout.
- Owns **AWS deployment with Terraform IaC and GitHub Actions CI/CD**; implements authentication, authorization, logging, monitoring, and error handling to meet enterprise security and auditability requirements.

### Entrata - Multifamily & Affordable Housing SaaS

Oct 2020 -- Jul 2024

**Senior Software Engineer** - 100K+ monthly active users · Python, React, AWS, Terraform

- **Served as primary technical contact for external vendor integrations**: worked directly with HUD and Tax Credit federal data source teams to scope requirements, debug API and data issues, coordinate go-live timelines, and resolve production incidents - led triage calls and wrote root cause summaries for non-technical stakeholders including PMs and TPMs.
- Integrated **enterprise systems via REST APIs**: external federal data sources, RabbitMQ event streams, and third-party platforms using versioned contracts and non-invasive integration patterns.
- Owned **PHI document infrastructure**: AWS S3 migration with Libsodium encryption, HIPAA-compliant access controls, audit logging - meeting enterprise security and compliance requirements in a regulated environment.
- Owned **Terraform IaC and GitHub Actions CI/CD**: Docker microservices (50% better scalability, 30% faster deployments); full environment promotion and secret management on AWS (EC2, RDS, S3, Lambda, SQS, IAM).
- Managed 20+ concurrent workstreams across 120+ Agile sprints; built Python/Bash automation (80% manual work reduction); reviewed 300+ PRs; mentored junior engineers.

### Entrata India

Sep 2018 -- Oct 2020

**Junior Software Engineer** - **Pune, India**

- Built production Python/PHP/PostgreSQL modules for regulated housing workflows; led AWS S3 cloud migration from legacy storage.

## PROJECTS

### Leasel - Production-Grade 8-Agent AI System

Python · LangGraph · FastAPI · RAG · Human-in-the-Loop · Agent Observability

End-to-end AI agent system for housing lease compliance: **tool execution, state management, orchestration** (LangGraph), RAG retrieval, and **human-in-the-loop escalation** for consequential decisions - the LLM never makes final eligibility or financial calls. Full evaluation harness (LLM-as-judge + golden datasets): 93.6% precision, 87.9% F1 across 1,240 test cases. Provenance on all material outputs; confidence scoring gates autonomy level. Demonstrates production-grade GenAI agent engineering with enterprise-level controls.

github.com/venkatalakshmiparimala/Lease_complaince_rag

### AgentProbe - LLMOps Evaluation & Observability Toolkit

Python · FastAPI · 5 LLM Providers · LLMOps · CI/CD

15-endpoint FastAPI service with **agent observability and LLMOps** instrumentation across five LLM providers. Evaluates accuracy, hallucination rate, retrieval quality, and compliance with CI regression tests and golden datasets. Built as a reusable evaluation substrate - the same pattern Sedgwick needs across multiple AI build streams.

github.com/venkatalakshmiparimala/AgentProbe

## EDUCATION

**M.S. Data Analytics Engineering** — George Mason University, Virginia — May 2026 | GPA 3.87

**B.Tech Computer Science & Engineering** — JNTU Anantapur — 2017